Active-Learning-Assisted Phase Boundary Refinement for 1D Altermagnetic FFLO TSC

Auto-generated by `ml_phase.report_builder`

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1 Physical and Computational Motivation

The exact BdG solver is expensive on dense grids. We use active learning to prioritize exact calls near uncertain or boundary-like regions.

2 Warm-Start Dataset

Run ID: active_boundary_h100_v1
Warm-start samples: 21528
Current exact samples: 21528
Iterations recorded: 0

3 Feature Construction

Raw coordinates:

$$x = [k_B T/t,\ J_A/t].$$

Regression targets:

$$[\Delta_{\text{opt}}, q_{\text{opt}}, \eta, I_c^+, I_c^-].$$

4 Feature Classification and Phase Labels

Class / label	Definition
normal	$\Delta_{\text{opt}} < \Delta_\epsilon$
uniform_SC	$\Delta_{\text{opt}} \geq \Delta_\epsilon$ and $\lvert q_{\text{opt}}\rvert < q_\epsilon$
FFLO	$\Delta_{\text{opt}} \geq \Delta_\epsilon$ and $\lvert q_{\text{opt}}\rvert \geq q_\epsilon$
eta_pos	$\eta > 0$
eta_neg	$\eta < 0$
strong_diode	$\lvert\eta\rvert > \eta_{\text{strong}}$

Table 1: Label definitions used in the active-learning loop.

5 ML Surrogate and Classifier Models

- Regression model family: Torch MLP Ensemble

- Classification model family: Torch MLP Ensemble
- Ensemble size: 5
- Feature dimension: 2
- Regression target dimension: 5
- Validation split: 0.15

6 Uncertainty Quantification

Classification uncertainty:

$$U_c(x) = 1 - \max_k p_k(x).$$

Regression uncertainty:

$$U_r(x) = \mathrm{std}_{m\in\mathrm{ensemble}}\hat{y}^{(m)}(x).$$

7 Acquisition Function

$$S(x) = w_c U_c(x) + w_r U_r(x) + w_\Delta B_\Delta(x) + w_q B_q(x) + w_\eta B_\eta(x) + w_g G(x) + w_d D(x).$$

8 Exact BdG Oracle

Selected points are evaluated with the exact BdG solver via a rank-aware oracle wrapper. Output observables are appended to the exact dataset after shard merge.

9 Multi-H100 Partitioning and Supercomputer Execution

- Execution mode: hpc
- World size: 8
- Partition strategy: round_robin
- Points per iteration: 128

10 Evaluation Metrics

Metric	Value
RMSE(Δ_{opt})	N/A
RMSE(q_{opt})	N/A
RMSE(η)	N/A
RMSE(I_c^+)	N/A
RMSE(I_c^-)	N/A
Phase accuracy	N/A
Boundary F1	N/A
Estimated exact-call reduction	N/A

Table 2: Latest available metrics from the active-learning run.

11 Active-Learning Results

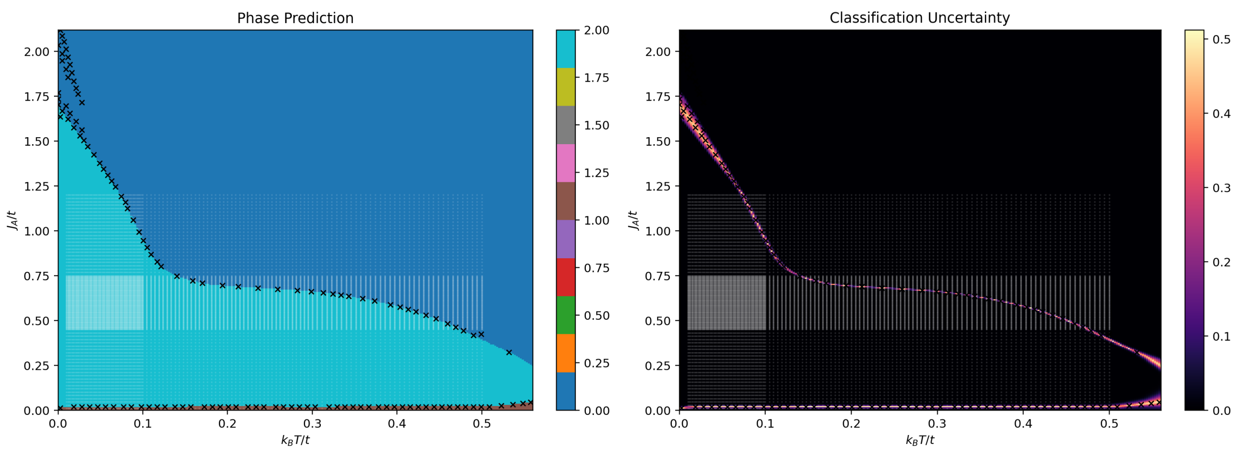

Figure 1: Phase prediction and uncertainty map (latest iteration).

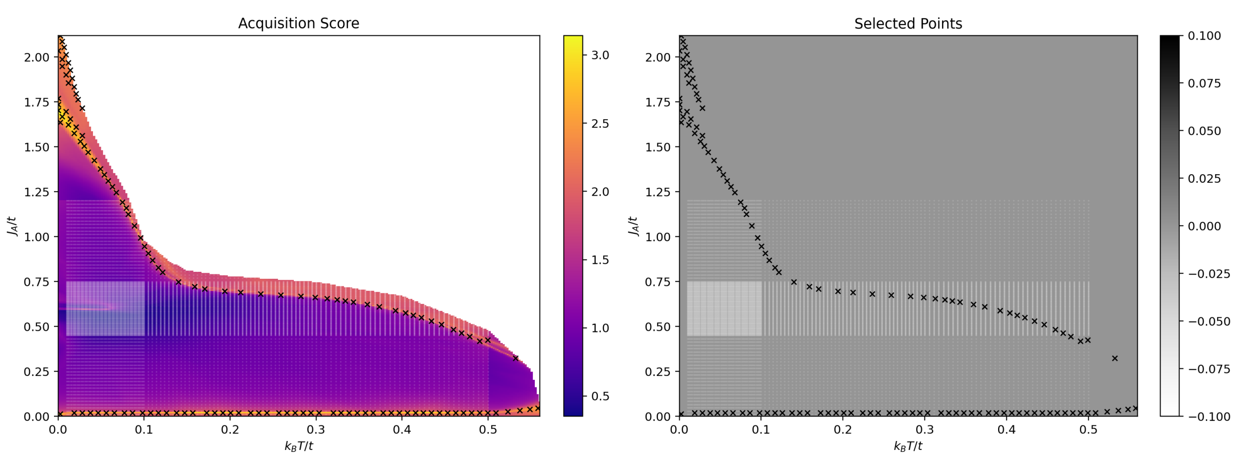

Figure 2: Acquisition score and selected points (latest iteration).

Figure 3: Learning-curve summary.

12 Discussion and Next Steps

- Extend labels with topological indicators (Z_2, OBC edge-mode checks).
- Add cost-aware multi-GPU scheduling if throughput measurements justify it.
- Increase point budget per iteration after H100 scaling tests.